Loumi TREMAS

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Professional Summary

Research Engineer and PhD candidate (defence expected 2026) specialising in **supervised deep learning for physical simulation** and generative models for inverse design. Full-stack research ownership: literature review, FDTD simulation pipeline, model training, benchmarking and deployment – end-to-end. Self-driven learner expanding into **self-supervised learning** (JEPA, Transformers, ViT) through hands-on reimplementations – see portfolio. Proven $10^6\times$ inference speedup over classical solvers at STMicroelectronics.

Technical Skills

ML & DL: PyTorch, JAX/Flax, CNNs (U-Net, ResNeXt), Generative Models (Diffusion, Schrödinger Bridges), Self-Supervised Learning (JEPA – from scratch), Transformers, ViT, HuggingFace.
Infrastructure: Python, CUDA, Distributed Training (Ray, multi-GPU), Azure, Linux, C.
Research: FDTD Simulation, Inverse Design, Surrogate Modelling, Benchmarking. **Languages:** French (Native), English (Professional)

Professional Experience

PhD Researcher – Deep Learning for Physical Simulation (CIFRE) — 2023 – 2026 (expected)
Paris-Saclay University × STMicroelectronics — Institut d'Optique Graduate School, Grenoble

- **Full-stack research ownership**: literature review, dataset design, FDTD simulation pipeline, model training, benchmarking and deployment – independently end-to-end.
- Trained supervised deep learning models (U-Net, ResNeXt) on electromagnetic simulation data – $10^6\times$ inference acceleration over classical FDTD solvers.
- Designed inverse design workflows combining gradient-based optimisation and Diffusion Schrödinger Bridges for photonic metasurface design; published at SPIE, SISPAD and META.
- Orchestrated distributed training on GPU clusters (Ray on Azure); end-to-end profiling and deployment.

AI Engineer — Aug 2022 – Feb 2023
STMicroelectronics — Crolles, France

- Built computer vision pipelines for automated anomaly detection in SEM images; hardened prototypes into production tools. Retained as CIFRE PhD researcher.

Education

PhD in AI & Computational Physics — 2023 – 2026 (expected)
Paris-Saclay University × STMicroelectronics — Institut d'Optique Graduate School
MSc Artificial Intelligence — 2022
Grenoble INP – Ensimag — Grenoble, France
BSc Computer Science — 2017 – 2021
Aix-Marseille University (Erasmus: Uppsala Universitet, Sweden) — France

ML Portfolio – Personal Projects

I-JEPA – from scratch — *PyTorch · SSL · RTX 3090*

- Reimplementation of *Image JEPA* (Assran et al., CVPR 2023): patch masking, dual-encoder EMA (τ=0.996), predictor Transformer. STL-10 (100k): loss $0.408 \to 0.088$ (-78%), 100 epochs, $\sim$72 min.

LeWorldModel – from scratch (arXiv:2603.19312) — *World Models · Robotics · AdaLN · SIGReg*

- Single ViT encoder, AdaLN action conditioning, SIGReg anti-collapse (no EMA), causal temporal masking. Dataset: `facebook/jepa-wms` pusht (224×224 RGB, 2-dim continuous actions).

Publications & Conferences

- Diffusion Schrödinger Bridges for Metasurface Inverse Design — arXiv:2602.06605, 2026
- Enhanced Posterior Sampling via Diffusion Models for Metasurface Inverse Design — arXiv:2601.15210, 2026
- Heuristic-Initialized Gradient Descent for Large-Scale Metasurface Inverse Design — SPIE, 2026
- Inverse Design of Optical Metasurface for CMOS Imagers: Multi-Objective Approach — SISPAD, 2025
- Transition to Neural Network Solutions for Flat Optical Devices — SISPAD, 2024
- *Self-Enhanced Generative Metasurface Inverse Design* — META, Dublin 2026
- *Impact of Numerical Simulation Boundary-Decomposition Errors* — META, Toyama 2024

Interests & Hobbies

Endurance Sports: Running and triathlon. **Automotive:** **4th place** at the **4L Trophy** (1 000+ teams) – 6 000 km humanitarian rally across the Moroccan desert.